Aim:Wireless Sensor Network Deployment and Coverage Analysis
Simulate random and grid deployment of sensor nodes and evaluate network coverage and connectivity.
CODE:

```
# ======================================================
# WSN Grid Deployment and Coverage Analysis
# NS2.35 Compatible
# ======================================================

set ns [new Simulator]

# Trace Files
set tracefile [open wsn.tr w]
$ns trace-all $tracefile

set namfile [open wsn.nam w]
$ns namtrace-all-wireless $namfile 500 500

# Topography
set topo [new Topography]
$topo load_flatgrid 500 500

# Simulation Parameters
set val(chan)       Channel/WirelessChannel
set val(prop)       Propagation/TwoRayGround
set val(netif)      Phy/WirelessPhy
set val(mac)        Mac/802_11
set val(ifq)        Queue/DropTail/PriQueue
set val(ll)         LL
set val(ant)        Antenna/OmniAntenna
set val(ifqlen)     50
set val(nn)         9
set val(rp)         AODV

create-god $val(nn)

# Configure Nodes
$ns node-config \
   -adhocRouting $val(rp) \
   -llType $val(ll) \
   -macType $val(mac) \
   -ifqType $val(ifq) \
   -ifqLen $val(ifqlen) \
   -antType $val(ant) \
   -propType $val(prop) \
   -phyType $val(netif) \
   -channelType $val(chan) \
   -topoInstance $topo \
   -agentTrace ON \
```

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```
    -routerTrace ON \
    -macTrace ON

# Create Nodes
for {set i 0} {$i < $val(nn)} {incr i} {
    set node(i)[ns node]
}

# ================================================
# GRID DEPLOYMENT (3x3)
# ================================================

$node(0) set X_ 100
$node(0) set Y_ 100

$node(1) set X_ 200
$node(1) set Y_ 100

$node(2) set X_ 300
$node(2) set Y_ 100

$node(3) set X_ 100
$node(3) set Y_ 200

$node(4) set X_ 200
$node(4) set Y_ 200

$node(5) set X_ 300
$node(5) set Y_ 200

$node(6) set X_ 100
$node(6) set Y_ 300

$node(7) set X_ 200
$node(7) set Y_ 300

$node(8) set X_ 300
$node(8) set Y_ 300

for {set i 0} {$i < $val(nn)} {incr i} {
    node(i) set Z_ 0
    $ns initial_node_pos node(i) 20
}

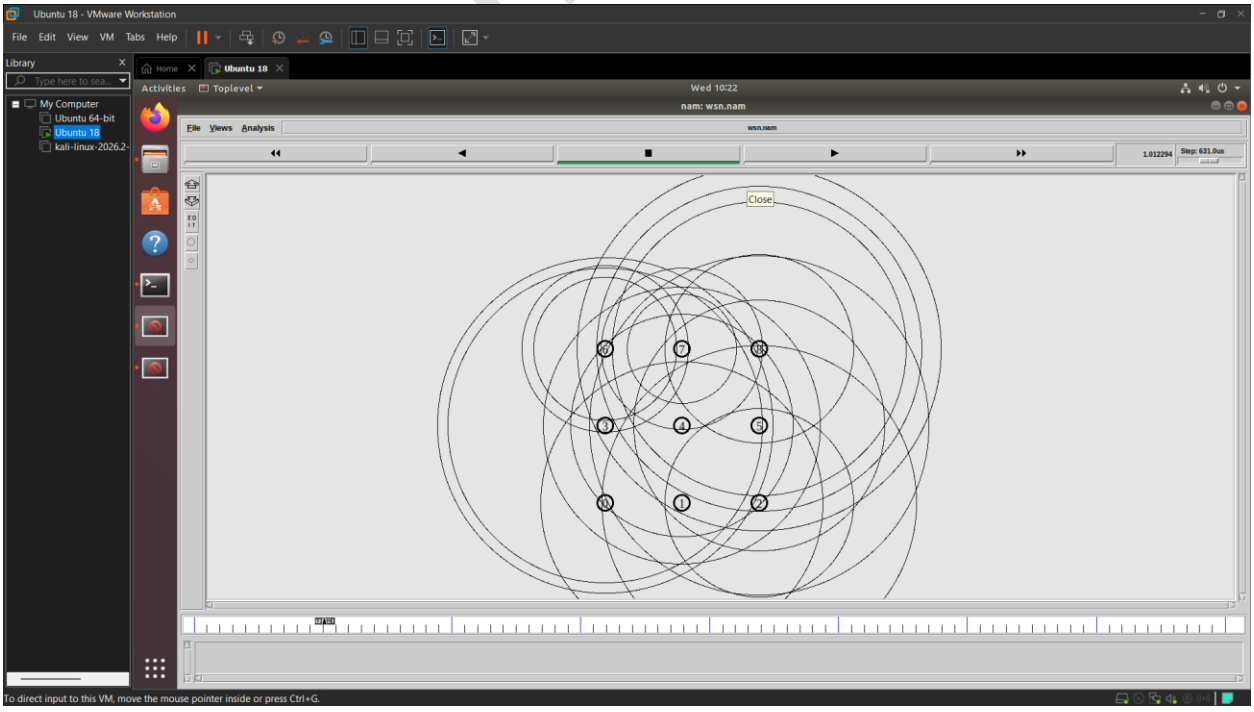
# ================================================
# Traffic
# ================================================

set udp0 [new Agent/UDP]
$ns attach-agent $node(0) $udp0
```

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```
set null0 [new Agent/Null]
$ns attach-agent $node(8) $null0
$ns connect $udp0 $null0
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1
$cbr0 attach-agent $udp0
nsat1.0"cbr0 start"
nsat8.0"cbr0 stop
# ======================================================
# Finish Procedure
# ======================================================
proc finish {} {
    global ns tracefile namfile
    $ns flush-trace
    close $tracefile
    close $namfile
    exec nam wsn.nam &
    exit 0
}
$ns at 10.0 "finish"
puts "Starting WSN Grid Deployment Simulation..."
$ns run
```

OUTPUT:



Observation

- Nine sensor nodes were deployed in a structured 3×3 grid pattern.
- Uniform spacing provided balanced network coverage.
- Node 0 acted as the source node and Node 8 as the destination.
- AODV successfully discovered routes between nodes.
- CBR traffic was transmitted without interruption.
- The grid topology maintained good connectivity among sensor nodes.
- Packet transmission was successfully visualized in NAM.

Result

The WSN grid deployment simulation was successfully executed. Sensor nodes were uniformly deployed, and data packets were transmitted successfully from the source node to the destination node.

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